

Hospital Management System

Functional Requirements

Functional Requirements

A. User and Role Management

FR-01: User Login

The system shall allow authorized users to log in using a username/email and password.

FR-02: Role-Based Access Control

The system shall restrict access based on user role.

Example:

- Receptionist can register patients and book appointments.
- Doctor can view patient medical history and write prescriptions.
- Lab technician can update lab results.
- Admin can manage users and departments.

FR-03: User Management

The administrator shall be able to create, update, deactivate, and assign roles to system users.

FR-04: Password Reset

The system shall allow users to reset their password securely.

B. Patient Management

FR-05: Patient Registration

The system shall allow the receptionist to register a new patient by entering personal, contact, insurance, and emergency contact information.

FR-06: Unique Patient ID

The system shall generate a unique patient ID for every registered patient.

FR-07: Patient Search

The system shall allow authorized users to search for patients by name, phone number, national ID, or patient ID.

FR-08: Patient Profile Update

The system shall allow authorized users to update patient information.

FR-09: Patient Medical History

The system shall store and display patient medical history, including previous visits, diagnoses, prescriptions, allergies, lab results, and admissions.

C. Appointment Management

FR-10: Appointment Booking

The system shall allow receptionists or patients to book appointments with doctors based on availability.

FR-11: Doctor Availability

The system shall display available appointment slots according to doctor schedule, department, and working hours.

FR-12: Appointment Rescheduling

The system shall allow authorized users to reschedule appointments.

FR-13: Appointment Cancellation

The system shall allow authorized users to cancel appointments and record the cancellation reason.

FR-14: Appointment Status Tracking

The system shall track appointment status:

- Scheduled
- Checked-in
- In consultation

- Completed
 - Cancelled
 - No-show
-

D. Consultation and Medical Records

FR-15: Patient Check-In

The system shall allow the receptionist or nurse to check in a patient when they arrive.

FR-16: Consultation Record

The system shall allow doctors to create a consultation record containing symptoms, diagnosis, notes, treatment plan, and follow-up instructions.

FR-17: Allergy and Risk Alerts

The system shall display alerts for allergies, chronic diseases, or critical medical notes before prescribing treatment.

FR-18: Prescription Creation

The system shall allow doctors to create prescriptions including medicine name, dosage, frequency, duration, and instructions.

FR-19: Medical Record Locking

The system shall prevent unauthorized editing of finalized consultation records.

E. Laboratory Management

FR-20: Lab Test Request

The system shall allow doctors to request lab tests for a patient.

FR-21: Lab Request Queue

The system shall allow lab technicians to view pending lab test requests.

FR-22: Lab Result Entry

The system shall allow lab technicians to enter or upload lab test results.

FR-23: Lab Result Review

The system shall allow doctors to review lab results related to their patients.

FR-24: Critical Result Alert

The system shall notify the doctor when a critical lab result is entered.

F. Pharmacy Management

FR-25: Prescription View

The system shall allow pharmacists to view approved prescriptions.

FR-26: Medicine Dispensing

The system shall allow pharmacists to mark medicines as dispensed.

FR-27: Medicine Stock Check

The system shall verify medicine availability before dispensing.

FR-28: Low Stock Alert

The system shall notify the pharmacist or administrator when medicine stock falls below the minimum threshold.

G. Admission and Discharge Management

FR-29: Patient Admission

The system shall allow authorized users to admit a patient to a department, room, and bed.

FR-30: Bed Availability

The system shall display available beds by department, room type, and status.

FR-31: Inpatient Monitoring

The system shall allow nurses and doctors to record inpatient notes, vital signs, treatments, and progress.

FR-32: Patient Transfer

The system shall allow authorized users to transfer a patient from one room, bed, or department to another.

FR-33: Discharge Process

The system shall allow doctors to initiate discharge and billing staff to finalize discharge after payment clearance.

H. Billing and Payment

FR-34: Bill Generation

The system shall generate bills based on consultations, lab tests, medicines, admission, procedures, and other hospital services.

FR-35: Insurance Handling

The system shall allow billing staff to apply insurance coverage if the patient has valid insurance.

FR-36: Payment Recording

The system shall allow cashiers to record full or partial payments.

FR-37: Invoice Generation

The system shall generate printable invoices and payment receipts.

FR-38: Outstanding Balance Tracking

The system shall track unpaid or partially paid bills.

I. Reporting and Administration

FR-39: Daily Appointment Report

The system shall generate daily appointment reports by doctor, department, or status.

FR-40: Revenue Report

The system shall generate revenue reports by date range, department, payment type, or service type.

FR-41: Patient Statistics

The system shall generate statistics about registered patients, visits, admissions, discharges, and common diagnoses.

FR-42: Audit Log

The system shall record important actions such as login attempts, patient data changes, prescription creation, billing updates, and medical record modifications.

Use Cases

Use Cases

UC-01: Register New Patient

Primary Actor

Receptionist

Goal

Create a new patient profile in the system.

Preconditions

The receptionist is logged in and has permission to register patients.

Main Flow

1. The receptionist opens the patient registration page.
2. The system displays the registration form.
3. The receptionist enters patient personal information.
4. The receptionist enters contact information.
5. The receptionist enters emergency contact information.
6. The receptionist enters insurance information, if available.
7. The receptionist submits the form.
8. The system validates the data.
9. The system creates a new patient profile.
10. The system generates a unique patient ID.
11. The system displays a confirmation message.

Alternative Flows

A1: Patient Already Exists

1. The system detects a possible duplicate patient using national ID, phone number, or date of birth.
2. The system displays matching patient records.
3. The receptionist reviews the existing records.
4. The receptionist either selects the existing patient or confirms creating a new profile.

A2: Missing Required Information

1. The receptionist submits the form with missing required fields.
2. The system highlights the missing fields.
3. The receptionist completes the required information.
4. The system continues registration.

Exception Flows

E1: System Fails to Generate Patient ID

1. The system fails to generate a unique patient ID.
 2. The system displays an error message.
 3. The registration is not completed.
 4. The system logs the error.
-

UC-02: Book Appointment

Primary Actor

Receptionist or Patient

Goal

Book an appointment with a doctor.

Preconditions

The patient exists in the system.

Main Flow

1. The actor opens the appointment booking page.
2. The actor searches for the patient.
3. The actor selects the department.
4. The system displays available doctors.
5. The actor selects a doctor.
6. The system displays available time slots.
7. The actor selects a suitable slot.
8. The system validates the slot availability.
9. The system books the appointment.
10. The system updates the slot status.
11. The system sends or displays confirmation.

Alternative Flows

A1: Doctor Has No Available Slots

1. The system shows no available slots for the selected doctor.
2. The actor selects another doctor or another date.

A2: Patient Requests Rescheduling

1. The actor selects an existing appointment.
2. The actor chooses a new available time slot.
3. The system updates the appointment.

Exception Flows

E1: Slot Becomes Unavailable During Booking

1. Another user books the same slot first.
 2. The system rejects the booking.
 3. The system asks the actor to select another slot.
-

UC-03: Check In Patient

Primary Actor

Receptionist or Nurse

Goal

Mark the patient as arrived for an appointment.

Preconditions

The patient has a scheduled appointment.

Main Flow

1. The actor opens today's appointment list.
2. The actor searches for the patient appointment.
3. The actor selects "Check In."
4. The system updates the appointment status to "Checked-in."
5. The system adds the patient to the doctor's waiting queue.

Alternative Flows

A1: Patient Arrives Late

1. The system detects that the appointment time has passed.
2. The actor confirms whether to allow check-in.
3. The system updates the appointment status if allowed.

Exception Flows

E1: Appointment Not Found

1. The patient does not appear in today's appointment list.
 2. The system allows the actor to search by patient ID.
 3. The actor may book a walk-in visit if permitted.
-

UC-04: Conduct Consultation

Primary Actor

Doctor

Goal

Record medical consultation details.

Preconditions

The doctor is logged in.

The patient is checked in.

Main Flow

1. The doctor opens the waiting queue.
2. The doctor selects the patient.
3. The system displays the patient profile and medical history.
4. The system displays allergy and risk alerts.
5. The doctor enters symptoms.
6. The doctor enters diagnosis.
7. The doctor writes consultation notes.
8. The doctor creates a treatment plan.
9. The doctor may create a prescription.

10. The doctor may request lab tests.
11. The doctor saves the consultation record.
12. The system updates the appointment status to “Completed.”

Alternative Flows

A1: Doctor Requests Lab Test Before Final Diagnosis

1. The doctor creates a lab request.
2. The consultation remains open.
3. The doctor reviews results later.
4. The doctor finalizes the diagnosis.

A2: Doctor Refers Patient to Another Department

1. The doctor selects referral.
2. The system creates a referral request.
3. The patient may book another appointment.

Exception Flows

E1: Medical Record Cannot Be Saved

1. The system fails to save the consultation.
 2. The system displays an error message.
 3. The doctor retries saving.
 4. The system logs the issue.
-

UC-05: Create Prescription

Primary Actor

Doctor

Goal

Create a prescription for a patient.

Preconditions

The patient has an active consultation.

Main Flow

1. The doctor opens the prescription section.
2. The doctor searches for medicine.
3. The system displays medicine information.
4. The doctor enters dosage, frequency, duration, and instructions.
5. The system checks allergy conflicts.
6. The doctor confirms the prescription.
7. The system saves the prescription.
8. The prescription becomes visible to the pharmacy.

Alternative Flows

A1: Allergy Conflict Detected

1. The system detects a conflict between medicine and patient allergy.
2. The system displays a warning.
3. The doctor changes the medicine or confirms with justification.

Exception Flows

E1: Medicine Not Found

1. The doctor searches for medicine that does not exist in the system.
 2. The doctor may enter a custom medicine name if permitted.
 3. The system marks it as non-formulary.
-

UC-06: Request Lab Test

Primary Actor

Doctor

Goal

Send a lab test request to the laboratory department.

Preconditions

The patient has an active consultation or admission record.

Main Flow

1. The doctor opens the lab request section.
2. The doctor selects one or more lab tests.
3. The doctor adds clinical notes if needed.
4. The doctor submits the lab request.
5. The system creates a lab request record.
6. The request appears in the lab technician queue.

Alternative Flows

A1: Urgent Lab Request

1. The doctor marks the request as urgent.
2. The system prioritizes it in the lab queue.
3. The system may notify the lab technician.

Exception Flows

E1: Selected Lab Test Is Unavailable

1. The system detects that a test is currently unavailable.
 2. The system informs the doctor.
 3. The doctor selects another test or cancels the request.
-

UC-07: Enter Lab Result

Primary Actor

Lab Technician

Goal

Record lab results for a patient.

Preconditions

A lab request exists.

Main Flow

1. The lab technician opens the lab request queue.
2. The lab technician selects a pending request.
3. The system displays patient and test details.

4. The technician enters result values or uploads a result file.
5. The technician submits the results.
6. The system validates the result format.
7. The system saves the result.
8. The system notifies the requesting doctor.

Alternative Flows

A1: Critical Result

1. The system detects that a result is outside the critical range.
2. The system marks the result as critical.
3. The system sends an urgent notification to the doctor.

Exception Flows

E1: Invalid Result Format

1. The technician enters an invalid value or file type.
 2. The system rejects the submission.
 3. The technician corrects the input.
-

UC-08: Admit Patient

Primary Actor

Receptionist, Nurse, or Admission Officer

Goal

Admit a patient to the hospital.

Preconditions

The patient exists in the system.
A doctor has recommended admission.

Main Flow

1. The actor opens the admission page.
2. The actor searches for the patient.
3. The actor selects the department.

4. The system displays available rooms and beds.
5. The actor selects a room and bed.
6. The actor enters admission reason.
7. The actor submits the admission.
8. The system creates an inpatient record.
9. The system marks the selected bed as occupied.

Alternative Flows

A1: No Available Bed

1. The system shows no available beds in the selected department.
2. The actor may select another department or add the patient to a waiting list.

Exception Flows

E1: Bed Becomes Occupied During Admission

1. Another user assigns the same bed first.
 2. The system rejects the admission.
 3. The actor selects another available bed.
-

UC-09: Generate Bill

Primary Actor

Billing Officer

Goal

Generate a patient bill for services received.

Preconditions

The patient has received one or more billable services.

Main Flow

1. The billing officer opens the billing module.
2. The billing officer searches for the patient.
3. The system retrieves all unpaid billable services.
4. The system calculates the total amount.

5. The system applies insurance coverage, if available.
6. The system displays the final bill.
7. The billing officer confirms the bill.
8. The system generates an invoice.

Alternative Flows

A1: Partial Insurance Coverage

1. The system calculates the covered amount.
2. The system calculates the patient's remaining balance.
3. The bill is generated with both values.

A2: Manual Discount

1. The billing officer applies a discount if authorized.
2. The system records the discount reason.
3. The system recalculates the bill.

Exception Flows

E1: Insurance Validation Fails

1. The system cannot validate the insurance information.
 2. The system marks the bill as pending insurance review.
 3. The billing officer may proceed with self-payment if approved.
-

UC-10: Discharge Patient

Primary Actor

Doctor and Billing Officer

Goal

Complete patient discharge from the hospital.

Preconditions

The patient is currently admitted.

Main Flow

1. The doctor opens the inpatient record.
2. The doctor enters discharge summary and instructions.
3. The doctor initiates discharge.
4. The system notifies billing.
5. The billing officer generates the final bill.
6. The patient completes payment or insurance clearance.
7. The billing officer confirms payment clearance.
8. The system marks the patient as discharged.
9. The system marks the bed as available.

Alternative Flows

A1: Pending Payment

1. The bill is not fully paid.
2. The system keeps discharge status as pending clearance.
3. The patient is discharged only after financial approval or payment.

Exception Flows

E1: Doctor Tries to Discharge Without Summary

1. The doctor submits discharge without required summary.
2. The system blocks the action.
3. The doctor completes the discharge summary.

Edge Cases

Edge Cases and Exception Scenarios

A. Patient Management Edge Cases

Edge Case	Description	Expected System Behavior
Duplicate patient registration	Same patient is registered twice using slightly different names	System should detect possible duplicates using national ID, phone, DOB, and name similarity
Missing national ID	Patient does not have national ID or passport	System should allow registration with temporary ID if hospital policy allows
Emergency patient with unknown identity	Patient arrives unconscious or without documents	System should create a temporary emergency profile
Name spelling differences	Patient name written differently in different visits	System should allow search using phone, ID, DOB, or patient number
Patient requests data correction	Patient wants to correct incorrect profile data	System should allow authorized update and keep audit history

B. Appointment Edge Cases

Edge Case	Description	Expected System Behavior
Double booking	Two users try to book the same slot at the same time	System should allow only the first successful booking
Doctor cancels schedule	Doctor becomes unavailable after appointments are booked	System should notify affected patients and allow rescheduling
Patient arrives without appointment	Walk-in patient arrives	System should allow walk-in visit if permitted
Patient is late	Patient arrives after appointment time	System should allow check-in based on hospital policy
No-show patient	Patient does not attend	System should mark appointment as no-show after defined time

C. Consultation Edge Cases

Edge Case	Description	Expected System Behavior
Doctor forgets to save consultation	Consultation page is closed accidentally	System should autosave draft if possible
Doctor prescribes medicine with allergy conflict	Medicine conflicts with patient allergy	System should display strong warning
Multiple doctors access same record	Two doctors open the same patient record	System should manage editing permissions and prevent overwriting
Consultation needs correction after finalization	Doctor notices mistake after saving	System should allow amendment with audit trail, not silent editing
Patient has critical medical alert	Patient has serious allergy or condition	System should display alert prominently

D. Laboratory Edge Cases

Edge Case	Description	Expected System Behavior
Lab result entered for wrong patient	Technician selects wrong request	System should show patient identity clearly before submission
Critical lab result	Result value is dangerous	System should notify doctor immediately
Result file upload fails	PDF or image fails to upload	System should show error and allow retry
Invalid result value	Text entered instead of numeric result	System should validate input based on test type
Lab test cancelled	Test cannot be performed	System should allow cancellation with reason

E. Pharmacy Edge Cases

Edge Case	Description	Expected System Behavior
-----------	-------------	--------------------------

Medicine out of stock	Prescribed medicine is unavailable	System should notify pharmacist and doctor
Partial dispensing	Only part of prescription is available	System should record partial dispensing
Expired medicine	Medicine exists but is expired	System should block dispensing
Duplicate dispensing	Pharmacist tries to dispense same prescription twice	System should prevent duplicate dispensing
Prescription changed by doctor	Doctor modifies prescription before dispensing	System should show latest approved prescription only

F. Admission and Bed Management Edge Cases

Edge Case	Description	Expected System Behavior
Same bed assigned twice	Two admission officers select same bed	System should lock bed once assigned
Emergency admission with no bed	No bed is available	System should allow waiting list or temporary emergency admission
Patient transfer during admission	Patient moves to another department	System should update bed status and transfer history
Discharge without payment clearance	Patient is medically ready but payment is pending	System should keep discharge pending or require admin approval
Bed status not updated	Patient discharged but bed remains occupied	System should automatically release bed after discharge confirmation

G. Billing Edge Cases

Edge Case	Description	Expected System Behavior
Insurance expired	Insurance is entered but no longer valid	System should reject coverage or mark for review
Partial payment	Patient pays part of the bill	System should track remaining balance

Service added after invoice generation	Additional service appears after bill is generated	System should allow invoice adjustment or new invoice
Unauthorized discount	User applies discount without permission	System should block action or require approval
Refund required	Patient overpaid or service cancelled	System should support refund workflow with approval

H. Security and Audit Edge Cases

Edge Case	Description	Expected System Behavior
Unauthorized access	User tries to open restricted patient data	System should deny access and log attempt
Failed login attempts	User enters wrong password multiple times	System should lock or temporarily block account
Sensitive data modification	User changes medical or billing data	System should record who changed what and when
Deleted user account	Staff member leaves hospital	System should deactivate account, not delete historical actions
Session timeout	User leaves system open	System should log out after inactivity

ER Diagram

Hospital Management System — ER Diagram Tables

This version presents the ER diagram as **database tables**, showing:

- Table name
 - Attributes
 - Primary keys
 - Foreign keys
 - Short description of each attribute
 - Relationships at the end
-

1. Role Table

Stores the different system roles, such as Admin, Doctor, Nurse, Receptionist, Lab Technician, Pharmacist, and Cashier.

Attribute	Key	Description
RoleID	PK	Unique identifier for each role.
RoleName		Name of the role, such as Admin, Doctor, Nurse, or Receptionist.
Description		Short explanation of what the role can do in the system.
IsActive		Indicates whether the role is currently active.

2. User Table

Stores login accounts for all hospital system users.

Attribute	Key	Description
UserID	PK	Unique identifier for each system user.

RoleID	FK	References the role assigned to the user.
FullName		Full name of the user.
Email		User email used for login and communication.
PasswordHash		Encrypted password value.
PhoneNumber		User contact number.
IsActive		Indicates whether the user account is active.
CreatedAt		Date and time when the user account was created.

3. Department Table

Stores hospital departments, such as Cardiology, Pediatrics, Emergency, Laboratory, and Radiology.

Attribute	Key	Description
DepartmentID	PK	Unique identifier for each department.
DepartmentName		Name of the hospital department.
Description		Short description of the department.
IsActive		Indicates whether the department is active.

4. Doctor Table

Stores doctor-specific information.

Attribute	Key	Description
DoctorID	PK	Unique identifier for each doctor.

UserID	FK	References the user account connected to this doctor.
DepartmentID	FK	References the department where the doctor works.
Specialty		Doctor's medical specialty.
LicenseNumber		Doctor's professional license number.
IsAvailable		Indicates whether the doctor is currently available for appointments.

5. InsuranceProvider Table

Stores insurance companies or providers linked to patients.

Attribute	Key	Description
InsuranceProviderID	PK	Unique identifier for each insurance provider.
ProviderName		Name of the insurance company.
ContactNumber		Insurance provider contact number.
Email		Insurance provider email address.
Address		Insurance provider address.
IsActive		Indicates whether this insurance provider is active.

6. Patient Table

Stores patient personal, contact, emergency, and medical profile information.

Attribute	Key	Description
PatientID	PK	Unique identifier for each patient.
InsuranceProviderID	FK, Nullable	References the patient's insurance provider, if available.

PatientCode	Unique readable patient code generated by the system.
FirstName	Patient's first name.
LastName	Patient's last name.
DateOfBirth	Patient's date of birth.
Gender	Patient's gender.
PhoneNumber	Patient's main contact number.
Address	Patient's home address.
NationalID	Patient's national ID or passport number.
EmergencyContactName	Name of the emergency contact person.
EmergencyContactPhone	Phone number of the emergency contact person.
BloodType	Patient's blood type, if known.
Allergies	Known patient allergies.
CreatedAt	Date and time when the patient record was created.
IsActive	Indicates whether the patient record is active.

7. Appointment Table

Stores appointments between patients and doctors.

Attribute	Key	Description
AppointmentID	PK	Unique identifier for each appointment.
PatientID	FK	References the patient who booked the appointment.
DoctorID	FK	References the doctor assigned to the appointment.
AppointmentDateTime		Date and time of the appointment.

Status		Current appointment status, such as Scheduled, CheckedIn, Completed, Cancelled, or NoShow.
Reason		Reason for the visit.
CancellationReason	Nullable	Reason for cancelling the appointment, if cancelled.
CreatedAt		Date and time when the appointment was created.

8. Consultation Table

Stores the medical consultation details written by the doctor.

Attribute	Key	Description
ConsultationID	PK	Unique identifier for each consultation.
PatientID	FK	References the patient receiving the consultation.
DoctorID	FK	References the doctor performing the consultation.
AppointmentID	FK, Nullable	References the appointment that resulted in this consultation.
ConsultationDate		Date and time of the consultation.
Symptoms		Symptoms reported by the patient.
Diagnosis		Doctor's diagnosis.
Notes		General medical notes written by the doctor.
TreatmentPlan		Treatment plan recommended by the doctor.
FollowUpInstructions		Instructions for patient follow-up.
IsFinalized		Indicates whether the consultation record has been finalized and locked.

9. Prescription Table

Stores prescriptions created during consultations.

Attribute	Key	Description
PrescriptionID	PK	Unique identifier for each prescription.
ConsultationID	FK	References the consultation where the prescription was created.
PrescriptionDate		Date and time when the prescription was created.
Notes		Additional notes about the prescription.
Status		Prescription status, such as Created, Approved, Dispensed, PartiallyDispensed, or Cancelled.

10. Medicine Table

Stores medicines available in the hospital pharmacy.

Attribute	Key	Description
MedicineID	PK	Unique identifier for each medicine.
MedicineName		Name of the medicine.
Category		Medicine category, such as antibiotic, painkiller, or vitamin.
QuantityInStock		Current available stock quantity.
MinimumStockLevel		Minimum quantity before the system sends a low-stock alert.
ExpiryDate		Expiration date of the medicine.
UnitPrice		Price of one unit of the medicine.
IsActive		Indicates whether the medicine is currently used in the system.

11. PrescriptionItem Table

Stores the medicines included in each prescription.

This table is important because one prescription can contain many medicines, and the same medicine can appear in many prescriptions.

Attribute	Key	Description
PrescriptionItemID	PK	Unique identifier for each prescription item.
PrescriptionID	FK	References the prescription.
MedicineID	FK	References the prescribed medicine.
Dosage		Medicine dosage, such as 500 mg.
Frequency		How often the medicine should be taken, such as twice daily.
Duration		Duration of treatment, such as 7 days.
Instructions		Special instructions for using the medicine.

12. LabTest Table

Stores the types of laboratory tests offered by the hospital.

Attribute	Key	Description
LabTestID	PK	Unique identifier for each lab test.
TestName		Name of the lab test.
TestCategory		Category of the test, such as Blood Test, Urine Test, or Biochemistry.
Price		Cost of the lab test.
NormalRange		Normal result range, if applicable.
IsAvailable		Indicates whether the lab test is currently available.

13. LabRequest Table

Stores lab test requests made by doctors.

Attribute	Key	Description
-----------	-----	-------------

LabRequestID	PK	Unique identifier for each lab request.
PatientID	FK	References the patient for whom the lab test was requested.
ConsultationID	FK	References the consultation that generated the lab request.
LabTestID	FK	References the requested lab test.
RequestDate		Date and time when the test was requested.
Priority		Priority level, such as Normal or Urgent.
Status		Request status, such as Pending, InProgress, Completed, or Cancelled.
ClinicalNotes		Notes from the doctor to the lab technician.

14. LabResult Table

Stores the result of a lab request.

Attribute	Key	Description
LabResultID	PK	Unique identifier for each lab result.
LabRequestID	FK	References the lab request related to this result.
ResultDate		Date and time when the result was entered.
ResultValue		Result value or text result.
ResultFilePath	Nullable	Path of uploaded result file, such as PDF or image.
Remarks		Additional comments from the lab technician.
IsCritical		Indicates whether the result is critical and requires urgent attention.

15. Room Table

Stores hospital rooms.

Attribute	Key	Description
RoomID	PK	Unique identifier for each room.
DepartmentID	FK	References the department where the room is located.
RoomNumber		Room number or room code.
RoomType		Type of room, such as Private, Shared, ICU, or Emergency.
IsActive		Indicates whether the room is available for use.

16. Bed Table

Stores beds inside hospital rooms.

Attribute	Key	Description
BedID	PK	Unique identifier for each bed.
RoomID	FK	References the room where the bed is located.
BedNumber		Bed number or bed code.
Status		Bed status, such as Available, Occupied, Reserved, or UnderMaintenance.

17. Admission Table

Stores inpatient admission records.

Attribute	Key	Description
AdmissionID	PK	Unique identifier for each admission.
PatientID	FK	References the admitted patient.
DoctorID	FK	References the supervising doctor.
BedID	FK	References the bed assigned to the patient.

AdmissionDate		Date and time when the patient was admitted.
DischargeDate	Nullable	Date and time when the patient was discharged.
AdmissionReason		Reason for admission.
Status		Admission status, such as Active, PendingDischarge, Discharged, or Transferred.
DischargeSummary	Nullable	Summary written when the patient is discharged.

18. Invoice Table

Stores patient bills.

Attribute	Key	Description
InvoiceID	PK	Unique identifier for each invoice.
PatientID	FK	References the patient receiving the invoice.
ConsultationID	FK, Nullable	References the consultation related to the invoice, if applicable.
AdmissionID	FK, Nullable	References the admission related to the invoice, if applicable.
InvoiceDate		Date and time when the invoice was generated.
TotalAmount		Total cost before insurance or payment.
InsuranceCoveredAmount		Amount covered by insurance.
PatientAmount		Amount the patient must pay.
Status		Invoice status, such as Unpaid, PartiallyPaid, Paid, Cancelled, or Refunded.

19. InvoiceItem Table

Stores the detailed services inside an invoice.

Attribute	Key	Description
InvoiceItemID	PK	Unique identifier for each invoice item.
InvoiceID	FK	References the invoice.
ServiceType		Type of billed service, such as Consultation, LabTest, Medicine, Admission, or Procedure.
Description		Description of the billed item.
Quantity		Number of units charged.
UnitPrice		Price per unit.
TotalPrice		Total price for this item. Usually $\text{Quantity} \times \text{UnitPrice}$.

20. Payment Table

Stores payments made against invoices.

Attribute	Key	Description
PaymentID	PK	Unique identifier for each payment.
InvoiceID	FK	References the invoice being paid.
PaymentDate		Date and time of the payment.
AmountPaid		Amount paid by the patient or insurance.
PaymentMethod		Payment method, such as Cash, Card, Bank Transfer, or Insurance.
TransactionReference	Nullable	Reference number for card, bank, or insurance transaction.
Status		Payment status, such as Completed, Pending, Failed, or Refunded.

21. AuditLog Table

Stores important system actions for security and traceability.

Attribute	Key	Description
AuditLogID	PK	Unique identifier for each audit log record.
UserID	FK	References the user who performed the action.
Action		Action performed, such as Create, Update, Delete, Login, or View.
EntityName		Name of the affected table or entity.
EntityID		ID of the affected record.
ActionDate		Date and time when the action occurred.
Details		Additional details about the action.

Relationships

1. One-to-Many Relationships

Role → User

One role can be assigned to many users.

Role 1 — many User

Example:

The Doctor role can be assigned to many doctor users.

Department → Doctor

One department can have many doctors.

Department 1 — many Doctor

Example:

The Cardiology department can include many doctors.

Department → Room

One department can contain many rooms.

Department 1 — many Room

Example:

The Emergency department can contain several rooms.

Room → Bed

One room can contain many beds.

Room 1 — many Bed

Example:

Room 205 can contain Bed 1, Bed 2, and Bed 3.

InsuranceProvider → Patient

One insurance provider can cover many patients.

InsuranceProvider 1 — many Patient

Example:

One insurance company can cover many registered patients.

Patient → Appointment

One patient can have many appointments.

Patient 1 — many Appointment

Example:

A patient may visit the hospital many times.

Doctor → Appointment

One doctor can receive many appointments.

Doctor 1 — many Appointment

Example:

A doctor can have several appointments in one day.

Patient → Consultation

One patient can have many consultations.

Patient 1 — many Consultation

Example:

A patient may have several consultations over time.

Doctor → Consultation

One doctor can perform many consultations.

Doctor 1 — many Consultation

Example:

A doctor can consult many patients.

Consultation → Prescription

One consultation can generate many prescriptions.

Consultation 1 — many Prescription

Example:

A doctor may create more than one prescription during a consultation.

Consultation → LabRequest

One consultation can generate many lab requests.

Consultation 1 — many LabRequest

Example:

A doctor may request a blood test and a urine test during the same consultation.

LabTest → LabRequest

One lab test type can appear in many lab requests.

LabTest 1 — many LabRequest

Example:

The CBC test can be requested for many patients.

Patient → LabRequest

One patient can have many lab requests.

Patient 1 — many LabRequest

Example:

A patient may need several lab tests.

LabRequest → LabResult

One lab request can produce one lab result.

LabRequest 1 — 0 or 1 LabResult

A lab request may not have a result yet if it is still pending.

Patient → Admission

One patient can have many admissions.

Patient 1 — many Admission

Example:

A patient may be admitted several times in different years.

Doctor → Admission

One doctor can supervise many admissions.

Doctor 1 — many Admission

Example:

A doctor may supervise many admitted patients.

Bed → Admission

One bed can appear in many admission records historically.

Bed 1 — many Admission

Important rule:

A bed can have many admissions over time, but only **one active admission at a time**.

Patient → Invoice

One patient can have many invoices.

Patient 1 — many Invoice

Example:

A patient may receive one invoice for consultation and another for admission.

Consultation → Invoice

One consultation can generate many invoices, or one invoice depending on hospital policy.

Consultation 1 — many Invoice

For a simpler system, you may enforce:

Consultation 1 — 0 or 1 Invoice

Admission → Invoice

One admission can generate many invoices.

Admission 1 — many Invoice

Example:

An inpatient case may generate interim invoices and a final invoice.

Invoice → InvoiceItem

One invoice contains many invoice items.

Invoice 1 — many InvoiceItem

Example:

An invoice may include consultation fee, lab test fee, medicine cost, and room charge.

Invoice → Payment

One invoice can have many payments.

Invoice 1 — many Payment

Example:

A patient may pay part of the invoice first and complete the rest later.

User → AuditLog

One user can perform many logged actions.

User 1 — many AuditLog

Example:

A receptionist may create several patient records, and each action is logged.

2. One-to-One or Optional One-to-One Relationships

User → Doctor

One doctor should have one user account.

User 1 — 0 or 1 Doctor

Explanation:

Not every user is a doctor. A user may be a receptionist, admin, nurse, or cashier. But if the user is a doctor, the user has one related Doctor profile.

Appointment → Consultation

One appointment may result in one consultation.

Appointment 1 — 0 or 1 Consultation

Explanation:

An appointment may be cancelled or marked as no-show, so it may not produce a consultation.

3. Many-to-Many Relationships

Prescription ↔ Medicine

A prescription can contain many medicines.

A medicine can appear in many prescriptions.

This many-to-many relationship is solved using the `PrescriptionItem` table.

Prescription many — many Medicine

Implemented as:

Prescription 1 — many PrescriptionItem

4. Relationship Summary Table

Relationship	Type	Implemented Through
Role → User	One-to-Many	User.RoleID
Department → Doctor	One-to-Many	Doctor.DepartmentID
Department → Room	One-to-Many	Room.DepartmentID
Room → Bed	One-to-Many	Bed.RoomID
InsuranceProvider → Patient	One-to-Many	Patient.InsuranceProviderID
Patient → Appointment	One-to-Many	Appointment.PatientID
Doctor → Appointment	One-to-Many	Appointment.DoctorID
Patient → Consultation	One-to-Many	Consultation.PatientID
Doctor → Consultation	One-to-Many	Consultation.DoctorID
Appointment → Consultation	Optional One-to-One	Consultation.AppointmentID
Consultation → Prescription	One-to-Many	Prescription.ConsultationID
Prescription → PrescriptionItem	One-to-Many	PrescriptionItem.PrescriptionID
Medicine → PrescriptionItem	One-to-Many	PrescriptionItem.MedicineID
Consultation → LabRequest	One-to-Many	LabRequest.ConsultationID
LabTest → LabRequest	One-to-Many	LabRequest.LabTestID
LabRequest → LabResult	Optional One-to-One	LabResult.LabRequestID
Patient → Admission	One-to-Many	Admission.PatientID
Doctor → Admission	One-to-Many	Admission.DoctorID
Bed → Admission	One-to-Many	Admission.BedID
Patient → Invoice	One-to-Many	Invoice.PatientID
Invoice → InvoiceItem	One-to-Many	InvoiceItem.InvoiceID

Invoice → Payment	One-to-Many	Payment.InvoiceID
User → AuditLog	One-to-Many	AuditLog.UserID
Prescription ↔ Medicine	Many-to-Many	PrescriptionItem

Front Design

Hospital Management System — Front-End Design

The front design should be organized around **modules**, because each hospital user has different responsibilities.

The main users are:

- Admin
 - Receptionist
 - Doctor
 - Nurse
 - Lab Technician
 - Pharmacist
 - Cashier / Billing Officer
-

1. General Layout

All system pages can use the same layout.

Main Layout Structure

Top Bar		
Hospital Logo	Search	Notifications User

Sidebar Menu	Main Content Area	
Dashboard	Page title	
Patients	Cards / Tables / Forms	
Appointments		
Consultations		
Laboratory		
Pharmacy		
Admissions		
Billing		
Reports		

2. Login Page

Purpose

Allows hospital staff to access the system securely.

Page Elements

Element	Description
Hospital Logo	Displayed at the top or left side.
Email / Username Field	User enters login email or username.
Password Field	User enters password.
Login Button	Submits login request.
Forgot Password Link	Allows password reset.
Error Message Area	Shows invalid login messages.

Suggested Design

|

Hospital Management System

|

|

[Hospital Logo]

|

|

Email / Username

|

[]

|

|

Password

|

[]

|

|

[Login]

|

3. Main Dashboard

Purpose

Shows a quick overview of hospital activity.

Dashboard Cards

Card	Description
Today's Appointments	Number of appointments scheduled today.
Waiting Patients	Patients currently checked in.
Available Doctors	Doctors available today.
Admitted Patients	Number of active admissions.
Pending Lab Requests	Lab tests waiting for results.
Unpaid Invoices	Bills not fully paid.

Suggested Design

Dashboard

[Today's Appointments] [Waiting Patients]

[Available Doctors] [Pending Lab Requests]

[Admitted Patients] [Unpaid Invoices]

Today's Appointment List

Patient | Doctor | Time | Status | Action

-----	-----	-----	-----	-----
Ali	Dr. X	9:00	Waiting	View

4. Patient Management Pages

4.1 Patient List Page

Purpose

Allows staff to search, view, and manage patients.

Page Elements

Element	Description
Search Bar	Search by patient name, phone, national ID, or patient code.
Add New Patient Button	Opens patient registration form.
Patient Table	Displays registered patients.
View Button	Opens patient profile.
Edit Button	Opens patient update form.

Suggested Design

Patients	[+ Add Patient]	
Search: [Name / Phone / Patient ID _____]		
Patient Code Name Phone Gender Actions		
-----	-----	-----
P-0001	Ali 03...	Male View Edit
P-0002	Sara 71...	Female View Edit

4.2 Add / Edit Patient Page

Purpose

Registers a new patient or updates existing patient data.

Form Sections

Section 1: Personal Information

Field	Type
First Name	Text
Last Name	Text
Date of Birth	Date
Gender	Dropdown
National ID / Passport	Text
Blood Type	Dropdown

Section 2: Contact Information

Field	Type
Phone Number	Text
Address	Text Area

Section 3: Emergency Contact

Field	Type
Emergency Contact Name	Text
Emergency Contact Phone	Text

Section 4: Insurance Information

Field	Type
-------	------

Insurance Dropdown
Provider

Insurance Number Text

Section 5: Medical Alerts

Field	Type
Allergies	Text Area
Chronic Diseases	Text Area

Suggested Design

| Add New Patient |

| Personal Information |

| First Name [] Last Name [] |

| DOB [] Gender [▼] Blood Type [▼] |

| National ID / Passport [] |

| Contact Information |

| Phone [] Address [] |

| Emergency Contact |

| Name [] Phone [] |

| Insurance |

| Provider [▼] Insurance Number [] |

| Medical Alerts |

| Allergies [] |

| Chronic Diseases [] |

| [Cancel] [Save] |

4.3 Patient Profile Page

Purpose

Shows all important patient details in one place.

Sections

Section	Description
Patient Summary	Basic data, patient code, age, phone.
Medical Alerts	Allergies, chronic diseases, blood type.
Appointments	Previous and upcoming appointments.
Consultations	Medical consultation history.
Prescriptions	Previous prescriptions.
Lab Results	Lab test history.
Invoices	Billing history.
Admissions	Admission history if applicable.

Suggested Design

Patient Profile: Ali Hassan	

Code: P-0001 Age: 35 Gender: Male	
Phone: 03... Insurance: Yes	

ALERTS	
Allergy: Penicillin	
Chronic Disease: Diabetes	

Tabs:

[Appointments] [Consultations] [Prescriptions]
[Lab Results] [Invoices] [Admissions]

Appointments	
Date Doctor Status Action	

5. Appointment Management Pages

5.1 Appointment List Page

Purpose

Displays all scheduled appointments.

Filters

Filter	Description
Date	Filter by appointment date.
Doctor	Filter by doctor.
Department	Filter by department.
Status	Scheduled, Checked-in, Completed, Cancelled, No-show.

Suggested Design

<

5.2 Book Appointment Page

Purpose

Books an appointment for a patient.

Form Fields

Field	Type
Patient	Search Select
Department	Dropdown
Doctor	Dropdown
Date	Date Picker
Available Time Slot	Button/List
Reason	Text Area

Suggested Design

| Book Appointment |

| Patient Search [] |

| Department [▼] |

| Doctor [▼] |

| Date [] |

| Available Slots |

| [09:00] [09:30] [10:00] [10:30] |

| Reason |

| [] |

| [Cancel] [Book] |

6. Doctor Consultation Pages

6.1 Doctor Dashboard

Purpose

Shows the doctor's daily work.

Elements

Element	Description
Today's Appointments	List of doctor appointments.
Waiting Queue	Checked-in patients waiting for consultation.
Pending Lab Results	Results waiting for doctor review.
Recent Consultations	Recently completed consultations.

Suggested Design

Doctor Dashboard			
[Today's Appointments] [Waiting Queue]			
[Pending Lab Results] [Completed Consultations]			
Waiting Queue			
Patient Time Reason Action			
----- ----- ----- -----			
Sara 10:00 Fever Start Consultation			

6.2 Consultation Page

Purpose

Allows the doctor to perform and record a consultation.

Page Sections

Patient Header

Shows patient identity and alerts.

Patient: Sara Ahmad | Age: 22 | Blood Type: O+
ALERT: Allergy to Penicillin

Consultation Form

Field	Type
Symptoms	Text Area
Diagnosis	Text Area
Notes	Text Area
Treatment Plan	Text Area
Follow-up Instructions	Text Area

Actions

Action	Description
Add Prescription	Opens prescription section.
Request Lab Test	Opens lab request form.
Save Draft	Saves consultation without finalizing.
Finalize Consultation	Locks consultation record.

Suggested Design

Consultation - Sara Ahmad

Age: 22 | Blood Type: O+ | Allergy: Penicillin

Symptoms

Diagnosis

Notes

Treatment Plan

| Follow-up Instructions |
| [] |

| [Add Prescription] [Request Lab Test] |
[Save Draft] [Finalize]

7. Prescription Page / Section

Purpose

Allows the doctor to add medicines to a prescription.

Fields

Field	Type
Medicine	Search Select
Dosage	Text
Frequency	Text
Duration	Text
Instructions	Text Area

Suggested Design

Prescription
Medicine [Search medicine...]
Dosage [] Frequency []
Duration []
Instructions []
[Add Medicine]

| Medicine | Dosage | Frequency | Duration | Remove |

```
|-----|-----|-----|-----|-----|
| Panadol | 500mg | 3 times | 5 days | X |
|-----|
| [Save Prescription] |
|-----|
```

8. Laboratory Pages

8.1 Lab Request Queue

Purpose

Allows lab technicians to see pending lab requests.

Suggested Design

```
|-----|
| Laboratory Requests |
|-----|
| Status [▼] Priority [▼] Date [____] |
|-----|
| Patient | Test | Priority | Status | Action |
|-----|-----|-----|-----|-----|
| Ali | CBC | Urgent | Pending | Enter Result |
|-----|
```

8.2 Enter Lab Result Page

Purpose

Allows lab technicians to enter or upload lab results.

Fields

Field	Type
Patient Name	Read-only
Test Name	Read-only

Result Value Text / Number

Result File File Upload

Remarks Text Area

Is Critical Checkbox

Suggested Design

Enter Lab Result	
Patient: Ali Hassan	
Test: CBC	
Requested By: Dr. X	
Result Value []	
Upload Result File [Choose File]	
Remarks []	
Critical Result? []	
[Cancel] [Submit Result]	

9. Pharmacy Pages

9.1 Prescription Queue

Purpose

Shows prescriptions waiting to be dispensed.

Suggested Design

Pharmacy - Prescriptions				
Status [▼] Date []				
Patient Doctor Date Status Action				
----- ----- ----- ----- -----				

| Sara | Dr. X | Today | Approved | Dispense |

9.2 Dispense Medicine Page

Purpose

Allows pharmacist to dispense prescribed medicines.

| Dispense Prescription |

| Patient: Sara Ahmad |

| Doctor: Dr. X |

Medicine	Required	In Stock	Status
Panadol	1 box	20 boxes	Available
Amoxicillin	1 box	0 boxes	Out of Stock

| [Mark as Partially Dispensed] [Dispense All] |

10. Admission and Bed Management Pages

10.1 Bed Availability Page

Purpose

Shows room and bed availability.

| Bed Availability |

| Department [▼] Room Type [▼] Status [▼] |

Room	Bed	Type	Status	Action
------	-----	------	--------	--------

```
|-----|-----|-----|-----|
| 201 | B1 | ICU | Available | Assign      |
| 201 | B2 | ICU | Occupied | View Patient |
|-----|
```

10.2 Admit Patient Page

Fields

Field	Type
Patient	Search Select
Doctor	Dropdown
Department	Dropdown
Room	Dropdown
Bed	Dropdown
Admission Reason	Text Area

```
| Admit Patient |
|-----|
| Patient [Search patient...] |
| Supervising Doctor [▼] |
| Department [▼] |
| Room [▼] |
| Bed [▼] |
| Admission Reason [ ] |
|-----|
| [Cancel] [Admit Patient] |
|-----|
```

11. Billing Pages

11.1 Invoice List Page

Purpose

Displays patient invoices.

```
-----
| Billing / Invoices          [+ New Bill] |
-----
| Search Patient [          ] Status [▼]   |
-----
| Invoice No | Patient | Date | Amount | Status | Action |
|-----|-----|-----|-----|-----|-----|
| INV-001   | Ali     | Today| $50   | Unpaid | View   |
-----
```

11.2 Generate Invoice Page

Fields and Sections

Section	Description
Patient Search	Select patient.
Services	Consultation, lab tests, medicine, admission charges.
Insurance	Apply insurance coverage if available.
Summary	Shows total, discount, insurance amount, patient amount.

```
-----
| Generate Invoice          |
-----
| Patient [Search patient...] |
-----
| Services                  |
| [ ] Consultation Fee    $30 |
| [ ] CBC Test            $15 |
| [ ] Medicine            $10 |
-----
| Insurance Provider: [▼]   |
| Insurance Covered Amount: [ ] |
-----
| Total Amount: $55        |
| Patient Amount: $40      |
```

| [Cancel] [Generate Bill] |

11.3 Payment Page

Purpose

Records patient payment.

Payment	
---------	--

Invoice: INV-001	
Patient: Ali Hassan	
Total Amount: \$55	
Paid Amount: \$20	
Remaining: \$35	

Payment Amount		
Payment Method	Cash ▼	
Transaction Reference		

| [Cancel] [Record Payment] |

12. Reports Pages

Report Types

Report	Description
Appointment Report	Appointments by date, doctor, department, or status.
Patient Report	Registered patients by date, gender, or age group.
Revenue Report	Revenue by date, department, or payment method.
Lab Report	Lab requests and results by status.

Admission Report	Active admissions and discharge history.
Pharmacy Report	Medicine stock and low-stock medicines.

Suggested Design

| Reports |

| Report Type [▼] |

| From Date [] To Date [] |

| Department [▼] |

| Status [▼] |

| [Generate Report] [Export PDF] [Export Excel] |

| Report Results |

| Table / Chart Area |

13. Admin Settings Pages

Admin Pages

Page	Purpose
Manage Users	Create, update, deactivate users.
Manage Roles	Define user roles.
Manage Departments	Add or update departments.
Manage Doctors	Add doctor profiles and assign departments.
Manage Lab Tests	Add available lab tests and prices.
Manage Medicines	Add medicines, stock, and prices.

Manage Rooms and
Beds

Configure rooms and beds.

14. Role-Based Sidebar Menus

Admin Sidebar

Dashboard
Users
Roles
Departments
Doctors
Patients
Appointments
Laboratory
Pharmacy
Admissions
Billing
Reports
Settings
Audit Logs

Receptionist Sidebar

Dashboard
Patients
Appointments
Check-In
Admissions
Invoices

Doctor Sidebar

Dashboard
Today's Appointments
Waiting Queue
Consultations
Prescriptions
Lab Requests

Lab Results
Patient History

Nurse Sidebar

Dashboard
Patient Queue
Admissions
Vital Signs
Inpatient Notes
Discharge Requests

Lab Technician Sidebar

Dashboard
Lab Requests
Pending Tests
Completed Results
Critical Results

Pharmacist Sidebar

Dashboard
Prescription Queue
Dispense Medicine
Medicine Stock
Low Stock Alerts

Cashier Sidebar

Dashboard
Invoices
Payments
Outstanding Balances
Payment Reports

15. Suggested Color and UI Style

Color Palette

Element	Suggested Color
Primary Color	Medical blue
Secondary Color	Light green
Background	Light gray or white
Warning	Orange
Error / Critical	Red
Success	Green

UI Style

Use a clean hospital-style interface:

- White background
 - Blue sidebar
 - Rounded cards
 - Clear tables
 - Large action buttons
 - Status badges
 - Search bar at the top of list pages
 - Alerts for allergies, critical results, unpaid bills, and low stock
-

16. Important UI Components

Use these reusable components across the system:

Component	Used For
Search Box	Patients, doctors, medicines, invoices
Dropdown List	Department, doctor, status, role
Date Picker	Appointments, reports, admissions

Status Badge	Appointment, invoice, lab request, admission
Confirmation Modal	Delete, cancel, discharge, finalize
Alert Box	Allergy, critical result, low stock
Data Table	Lists and reports
Tabs	Patient profile sections
Cards	Dashboard summary
File Upload	Lab results and patient documen

Services

